Raza Mustafa

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Professional Summary

13+ years of professional experience in academia, international research, and industry-level software development. Specialized in AI, machine learning, NLP, 5G networks, multimedia systems, and data-driven applications. Experienced in curriculum design, program leadership, full-stack development, database systems, web automation, and AI-enabled software solutions.

Experience

Assistant Professor Aug 2024 – Present

Loyola University New Orleans, USA

- I teach courses such as Python, Web Technologies, Networks, and Business Analytics, and consistently strive to design innovative curricula that equip students for emerging technology trends
- Actively promote diversity and inclusion in STEM by supporting students from diverse backgrounds in both research and classroom environments

FELLOWSHIP & RESEARCH

- Awarded the Marquette Fellowship from Loyola University New Orleans to conduct research on a project titled "Mapping Hidden Meanings: LLMs Capabilities for Coded Tropes Discovery"
- Awarded Llorens Family Faculty and Staff Excellence Fellowship
- Mentored undergraduate students to publish peer-reviewed research at leading international conferences, including MSWiM 2025 and IEEE ICC 2026.

SERVICES

- Director of Computer Science program
- Serving on the Faculty Senate Artificial Intelligence Committee
- Committee on Diversity, Equity, and Inclusion
- AI Strategy Subcommittee
- Faculty advisor to undergraduate students
- Coordinate and oversee the Computer Science internship program, including student progress, Career Development Center collaboration, and final grading

CURRICULUM DEVELOPMENT

- Introduced COSC A225: Python & Data Science
- AI Minor (In-progress)

Postdoctoral Research Fellow Mar 2023 – July 2024

American University, USA

- Developed a system using large language models that detects coded hate speech and extracts new coded hate keywords from the text

Research Associate Jun. 2022 – Nov 2022

INRIA, Sophia Antipolis, France

- Machine/Deep Learning solutions for predicting the quality of experience of YouTube videos in 5G architecture
- Build classifiers that predict the quality of experience using only wireless 5G metrics in real time

Graduate Researcher Jun 2018 – Dec 2022
University of Campinas, Brazil

- Proposed two open-source frameworks for multimedia experiments in 5G and beyond networks, including protocols (TCP, QUIC) and video content in HTTP adaptive streaming over DASH
- We are the first to find features from Inter Packet Gap from the encrypted stream and later developed machine learning classifiers to predict the quality of experience from these metrics

Professional Lecturer Apr 2015 – May 2018
UOL/COMSATS, Pakistan

- Use industry experience of 3+ years to teach students advanced knowledge in technology development, especially in web apps, graphics, and mobile applications
- Supervised final year projects (15+), advised master's students' theses (5+), and worked as coordinator of the master's program

Full Stack Developer Aug 2012 – Mar 2015
Qualityclix, Remote (USA)

- Development of web application systems. Gain hands-on experience in PHP, HTML, CSS, and MySQL
- Design custom responsive website designs in Photoshop and PSD to HTML conversion
- Deployment of web applications live on servers, data management, and maintenance
- Hands-on experience in WordPress - Themes, Graphics, CSS, Navigation, etc
- Design custom crawlers to extract millions of public records from the internet

Education

University of Campinas, Brazil
PhD Computer Engineering Aug 2018 – Dec 2022

- Advisor: Christian Esteve Rothenberg
- Dissertation Topic: Machine Learning Assisted DASH Video QoE Inference Through Network QoS Features in 4G and 5G Scenarios

Honors and Awards

- Ericsson-funded Ph.D. Scholarship, University of Campinas, Brazil, 2018–2022.
- Research Exchange Scholarship (€9,000), INRIA Sophia Antipolis, France, 2022.

Technical Skills

- **Machine & Deep Learning**: Python, Scikit-Learn, PyTorch, TensorFlow, NumPy, Pandas; experience with Transformer-based architectures and LLMs (ChatGPT, GPT, BERT) for NLP tasks
- **Data & Business Analytics**: Matplotlib, Seaborn, Tableau, BeautifulSoup, SciPy, Plotly for data analysis, visualization, and dashboarding
- **Web Engineering**: PHP, JavaScript, HTML5, CSS3, Ajax, Bash, JQuery, Bootstrap
- **Database Management**: Oracle, SQL, MySQL, SQLite; schema design, optimization, and query processing
- **Creative Tools & Office Suite**: Adobe Photoshop, Adobe Premiere Pro, Microsoft Office Suite; experienced with multiple IDEs

Undergraduate Research Mentoring and Scholarly Outcomes

- Mentored undergraduate research resulting in the following peer-reviewed publications and submissions:
  - "Evaluating Large Language Models for Implicit Hate Speech Detection," IEEE International Conference on Communications (IEEE ICC 2026).
  - "Mobile 360° Video QoE: Empirical Analysis of 5G QoS Metrics," ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM 2025).
  - Manuscript submitted to the International Conference on Machine Learning and Applications (ICMLA 2026).
- Provide undergraduate researchers with hands-on experience across the complete research lifecycle, including problem formulation, literature review, experimental design, data analysis, machine learning, scientific writing, and preparation of research for publication.

Selected Publications

- "From Radio Signal Dynamics to QoE Degradation Prediction in Immersive Streaming", Raza Ul Mustafa, 17th International Conference on Network of the Future (NoF 2026), 29 September–02 October 2026, Rome, Italy.
- "Mobile 360° Video QoE: Empirical Analysis of 5G QoS Metrics", Raza Ul Mustafa, Roi Dupart, et al., The 27th International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems, 2025.
- "QoE of 2D and 360 Video: Insights from 5G Radio Metrics", Raza Ul Mustafa, Sesha Dassanayake, et al., Journal of Mobile Networks and Application, 2025.
- "Investigating the Impact of Channel Metrics in 5G NSA and SA on YouTube and 360 Video", Raza Ul Mustafa, Tariqul Islam, Christian Esteve Rothenberg, IEEE/IFIP Network Operations and Management Symposium 12–16 May 2025 USA.
- "Monitoring the Evolution of Antisemitic Hate Speech on Extremist Social Media", Raza Ul Mustafa and Nathalie Japkowicz, IEEE DPSH – Digital Platform and Societal Harms, 2024.
- "Can GPT-4 Detect Subcategories of Hatred?", Raza Ul Mustafa and Nathalie Japkowicz, IEEE DPSH – Digital Platform and Societal Harms, 2024.
- "Coded Term Discovery for Online Hate Speech Detection", Dhanush, Raza Ul Mustafa, et al., The 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA 2024).
- "YouTube goes 5G: QoE Benchmarking and ML-based Stall Prediction", Raza Ul Mustafa, and Chadi Barakat and Christian Rothenberg, 2024 Wireless Communications and Networking Conference WCNC, 2024.
- "EFFECTOR: DASH QoE and QoS Evaluation Framework For EnCrypTed videO tRaffic", Raza Ul Mustafa, Christian Rothenberg and Pedro Henrique Gomes, Md Tariqul Islam, IEEE/IFIP Network Operations and Management Symposium, 2023.
- "Machine Learning-Assisted Closed-Control Loops for Beyond 5G Multi-Domain Zero-Touch Networks", NFS de Sousa, MT Islam, Raza Ul Mustafa, DAL Perez, CE Rothenberg, Journal of Network and Systems Management 30 (3), 1-29, 2022.
- "Machine learning approach to estimate video QoE of encrypted dash traffic in 5G networks", Raza Ul Mustafa, D Moura, CE Rothenberg, 2021 IEEE Statistical Signal Processing Workshop (SSP), 586-589, 2021.
- "A supervised machine learning approach for dash video QoE prediction in 5G networks", Raza Ul Mustafa, S Ferlin, C Esteve Rothenberg, D Raca, J J. Quinlan, Proceedings of the 16th ACM Symposium on QoS and Security for Wireless and Mobile Networks, 2020.

- "DASH QoE performance evaluation framework with 5G datasets", Raza Ul Mustafa, MT Islam, C Rothenberg, S Ferlin, D Raca, JJ Quinlan, 2020 16th International Conference on Network and Service Management (CNSM), 1-6, 2020.
- "A multiclass depression detection in social media based on sentiment analysis", Raza Ul Mustafa, N Ashraf, FS Ahmed, J Ferzund, B Shahzad, A Gelbukh, 17th International Conference on Information Technology, 2020.
- "Predicting the cricket match outcome using crowd opinions on social networks: A comparative study of machine learning methods", Raza Ul Mustafa, MS Nawaz, MIU Lali, T Zia, W Mehmood, Malaysian Journal of Computer Science 30 (1), 63-76, 2017.
- "Finding healthcare issues with search engine queries and social network data", MIU Lali, Raza Ul Mustafa, K Saleem, MS Nawaz, T Zia, B Shahzad, International Journal on Semantic Web and Information Systems (IJSWIS), 2017.

Professional Projects

American University Signature Research Initiative Program, USA Mar 2023 – Jul 2024
- Monitoring coded language emergence and use on extremist social media platforms.

5G real-time QoE prediction – INRIA France Jun 2022 – Nov 2022
- Analyzed user behavior and video experience quality with alternative metrics due to the rise of 5G and increased YouTube traffic.

CLAML – Ericsson Brazil Jun 2018 – Dec 2022
- To increase end-user satisfaction, developed machine learning solutions to predict the quality of experience using highly correlated quality of service features (metrics).
- Deployed models in real-time to edge nodes, which passively monitor encrypted traffic and estimate experience.

5G services in programmable networks – Ericsson Brazil Jun 2018 – Dec 2022
- Developed ML solutions to predict user experience using correlated quality of service metrics.

Professional Service

- Technical Program Committee (TPC) Member, ECML PKDD 2025.
- Technical Program Committee (TPC) Member, IEEE Conference on Digital Platforms and Societal Harms (DPSH), 2024.
- Reviewer, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024.

Invited Talks and Workshops

- Invited Talk, "Unmasking Hate Speech from Social Media Using Transformer Architectures."
- Invited Speaker, 2nd International Conference on Recent Advances in Computer Science and Information Technology (RACSIT), 2024.
- Instructor, Python Programming Workshop for students.

AI Product Development

- Founder and developer of **YourScholar.ai**, an AI-powered academic platform for discovering professors, exploring research opportunities, and supporting research-focused applications.